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MPA-MLF - Final project

github link: https://github.com/Roman-Lunin/MPA-MLF-PROJECT

Classification of room occupancy

1. Introduction:

The goal of this project is to classify the number of people in a room using machine learning applied to delay-Doppler domain signal representations.

The dataset consists of 2D snapshots representing reflections of signals from objects (humans or machines). Based on these snapshots, the goal is to classify the number of people in the room correctly.

The classification task includes four classes:

- Class 0: Machine only (no people)
- Class 1: One person
- Class 2: Two people
- Class 3: Three people

The main objective is to make a robust classifier that achieves the highest possible accuracy on unseen test data while maintaining good performance across all classes.

2. Problem Description

The dataset consists of labeled training images and unlabeled test images. Each training sample contains an image ID and a corresponding class label (the right answer for each image).

And we saw that the dataset is imbalanced, as shown in Fig. 1. Classes 1 and 2 are more frequent, while class 3 appears significantly less often.

To address this, class weights were computed as inverse frequencies. This ensures that rare classes (especially class 3) have higher importance during training and are not ignored by the model.

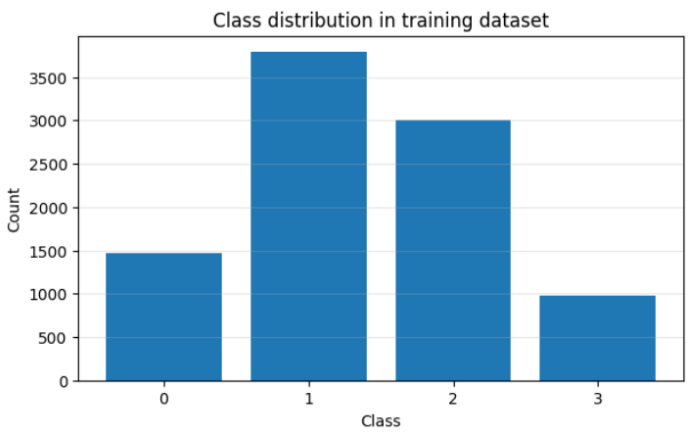

Fig. 1. Class distribution in the training dataset

Another important detail is that image filenames start from 1, while labels start from 0.

3. Data preprocessing

Image files were then collected from the extracted zip folders. In total, the dataset contained 9227 training images and 3955 test images.

Mapping

A very important preprocessing step was mapping CSV IDs to actual image files. According to the assignment, image filenames start from 1, while CSV IDs start from 0. Therefore, the correct image path was obtained using **file_num = id + 1**

id = 0 → img_1.png
id = 1 → img_2.png
id = 2 → img_3.png

After mapping, the dataframe contained three important columns:

id | target | image_path

Data Augmentation and Image Transformations

Before images were introduced into the model, several transformations were made.

For training data, the following transformations were used: resizing to **256 × 256(the best size for the chosen model)**, small random translation and scaling (move or zoom in/out), slight colour changes using ColorJitter, conversion to tensor, and normalization using ImageNet statistics.

The purpose of these transformations was to improve generalization. During training, every image can be slightly modified each time it is loaded. This means that the model does not always see the same image, but different small variations of it. As a result, the model is less likely to memorize training images and more likely to learn general visual patterns.

Test-time augmentation was also used. During prediction, each test image was evaluated several times: once normally and several times with small random transformations. The predicted probabilities were then averaged. This helped make the final predictions more stable.

Training Strategy of our model

The model was trained using **5-fold stratified cross-validation**.

The dataset was divided into five parts (Fold1, Fold2…Fold5). In each training round, four parts were used for training, and one part was used for validation. This means that each model was trained on approximately 80% of the data and validated on the remaining 20%. The process was repeated five times, so every part of the dataset was used once as validation data. This strategy was chosen because it gives a more reliable estimate of model performance than a single train-validation split. It also reduces dependence on one random data split.

Final Prediction and Test-Time Augmentation

After cross-validation training, predictions were generated for the test dataset. The final prediction pipeline used all saved checkpoints (2 models from each fold) and test-time augmentation.

For each checkpoint, the model predicted probabilities for all test images. When TTA was enabled, each image was predicted once normally and three more times with small random transformations. These probabilities were averaged. The final class was selected using the maximum average probability.

4. Model Architecture

For this project, we used **EfficientNet-B1**, implemented through the *timm* library. EfficientNet was selected because it is a strong convolutional neural network (CNN) architecture for image classification and provides a good balance between accuracy, speed, and memory usage.

The model was initialized with pretrained weights. This means that the model had already learned useful low-level image features, such as edges, shapes, and textures, from a large image dataset. We then fine-tuned it for our specific task with four output classes.

The final classification layer was adapted to output four values, corresponding to the four room occupancy classes.

Model configuration was:

- Input size: 256 × 256
- Batch size: 32
- Epochs: 14
- Learning rate: 0.0006
- Loss function: CrossEntropyLoss
- Label smoothing: 0.03
- Weight decay: 1e-4

5. Validation Results

After training and tuning, our model achieved 97,291% accuracy. After predicting labels for test data and submitting them to Kaggle, the actual score is 0.97667.

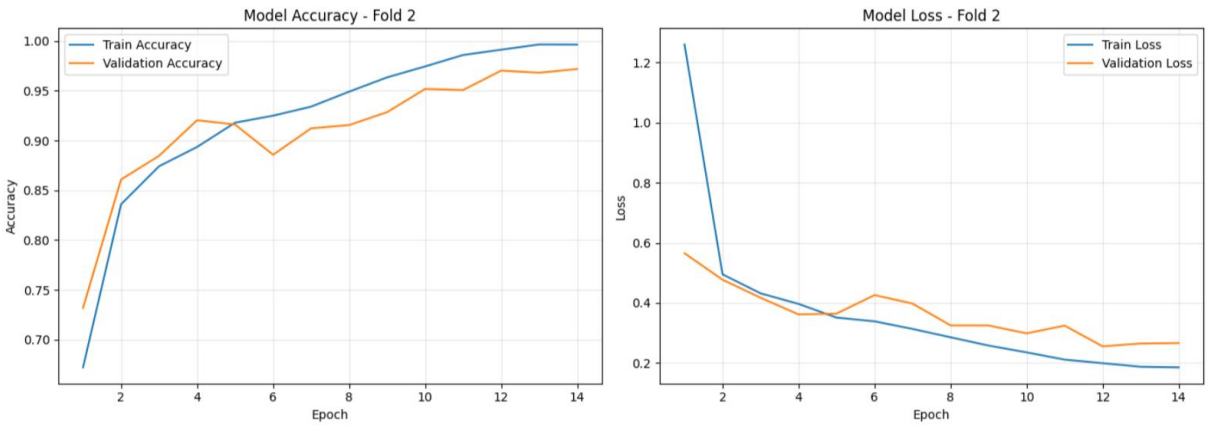

Fig. 2. Accuracy and loss graphs for the best model

Confusion Matrix Analysis

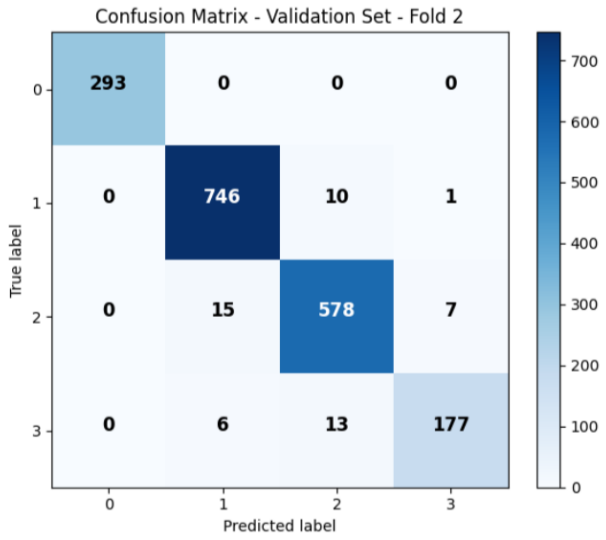

Fig. 3. Confusion matrix for the best model

Most errors occurred between neighbouring classes, especially between classes 1, 2, and 3. This is expected because distinguishing between two and three people can be more difficult when the signal reflections overlap or contain noise.

Test Prediction Distribution

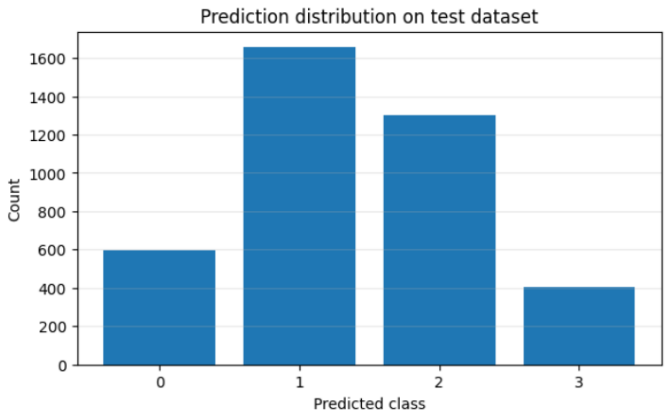

Fig. 4. Prediction distribution on the test dataset

The prediction distribution is not identical to the training class distribution, but it follows a similar general pattern: classes 1 and 2 appear most often, while class 3 appears less frequently. This suggests that the model does not collapse into predicting only one class and is able to distinguish between all four categories.